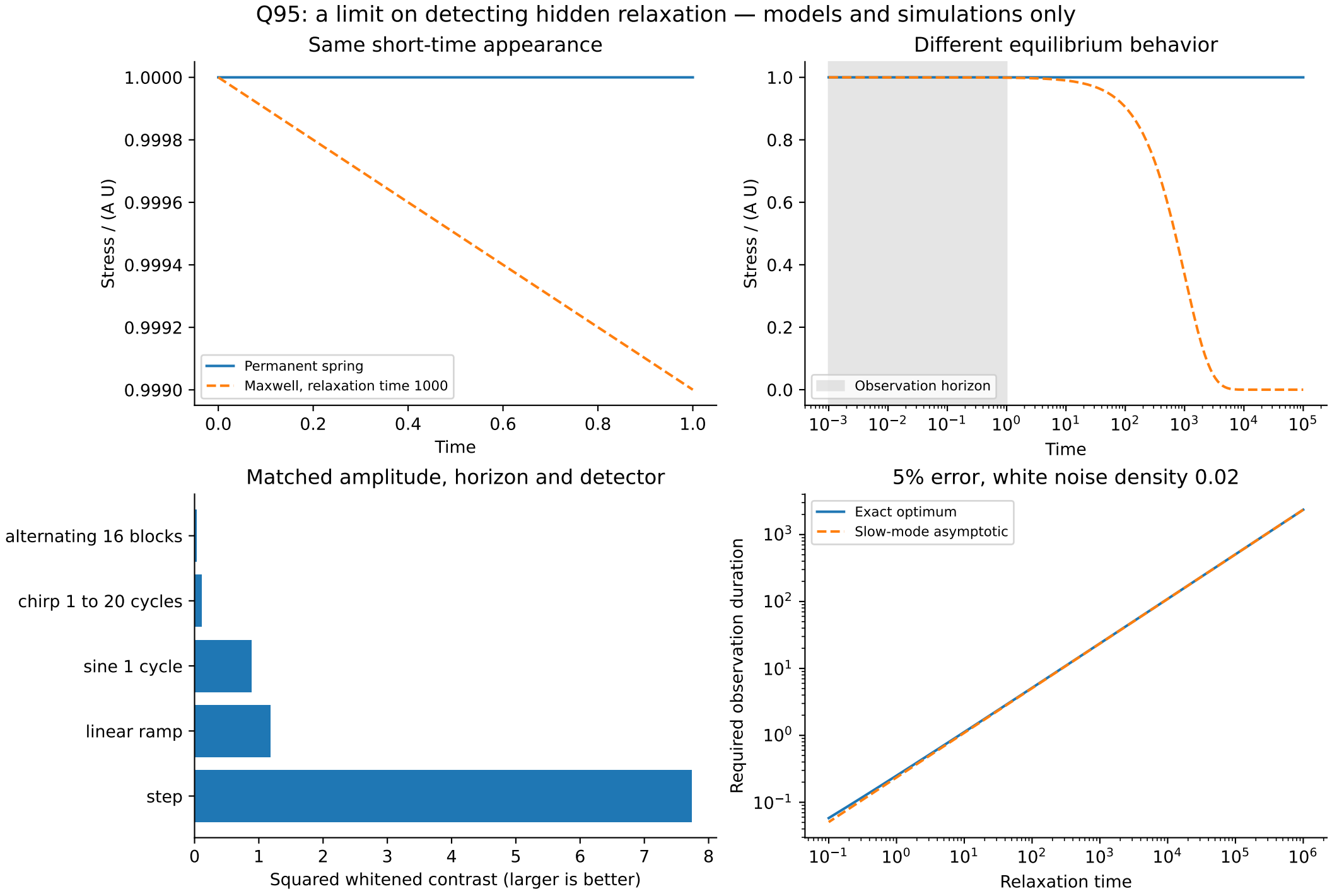
Q95: a limit on detecting hidden relaxation — models and simulations only
Same short-time appearance
Stress / (A U)
1.0000
0.9998
0.9996
0.9994
0.9992
0.9990
0.0
0.2
0.4
0.6
0.8
1.0
Time
Permanent spring
Maxwell, relaxation time 1000
Different equilibrium behavior
Stress / (A U)
1.0
0.8
0.6
0.4
0.2
0.0
10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5}
Time
Observation horizon
Matched amplitude, horizon and detector
alternating 16 blocks
chirp 1 to 20 cycles
sine 1 cycle
linear ramp
step
0 1 2 3 4 5 6 7 8
Squared whitened contrast (larger is better)
5% error, white noise density 0.02
Exact optimum
Slow-mode asymptotic
Required observation duration
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-1} 10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5} 10^{6}
Relaxation time